

Phloroglucinol-HCl Staining

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Purpose. Lignin visualization in whole seedpods or sections.

Reagent / condition	Preparation or use
3% phloroglucinol in ethanol	300 mg / 10 mL absolute ethanol; or 1.5 g / 50 mL
Whole seedpods	3% phloroglucinol, 37 °C, 3 days; replace solution daily; then 37% HCl for 5 min
Sections	Apply saturated phloroglucinol in 20% HCl

1. Prepare fresh phloroglucinol solution in absolute ethanol.
2. For seedpods, incubate at 37 °C for 3 days, replacing the staining solution daily.
3. Apply 37% HCl for 5 min.
4. Mount in ethanol and image with a stereomicroscope.